# XIAMEN UNIVERSITY MALAYSIA

## MID-TERM EXAMINATION

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| Course Code: | COM210 | |
| Course Name: | Engineering Maths III | |
| Question paper setter: | Georgios Fotopoulos | |
| Academic Session: | 2022-09 | |
| Total No. of Pages: | 3 | Time Allocated: 2 hours |

## INSTRUCTIONS TO CANDIDATES

1. This paper consists of 3 questions. Answer all questions.
2. Please follow the requirement of each question and give the corresponding answers in **PDF format**.
3. For every problem/subproblem, around **20%** of the total points will be allocated for presentation of solutions (clarity, coherence, conciseness, organization).
4. Show your work and organize your solution. **Answer without proper justification may receive less, or even zero credit.**
5. Communication between candidates in any means is forbidden. Answers must be entirely individual candidate's independent effort. If you are found sharing your solutions with other candidates, or suspected of doing so, you would be penalized accordingly.

(Student ID:_____ Full Name: _____)

**Question 1 (35 points).**

a. Find a geometric power series, centered at 0, for the function [5]

$$f(x) = \frac{2}{3-x}.$$

b. Find a power series for the function, centered at $c$, and determine the radius of convergence. [5]

$$g(x) = \frac{6}{4-x}, \quad c = 1.$$

c. A curve is defined by the parametric equations

$$x = \tan^2\theta, \quad y = \sec\theta, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

Eliminate the parameter to find a Cartesian equation of the curve. [5]

d. Given $\mathbf{r}(t) = \langle\sqrt{2-t}, (e^t-1)/t, \ln(t+1)\rangle$, find [5]

i. the domain of $\mathbf{r}(t)$;

ii. $\lim_{t\to 0} \mathbf{r}(t)$.

e. Find a vector function that represents the curve of intersection of the two surfaces, the cylinder $x^2 + y^2 = 4$ and the surface $z = \frac{xy}{4}$. [5]

f. The curves $\mathbf{r}_1(t) = \langle t, t^2, t^3\rangle$, $\mathbf{r}_2(t) = \langle\sin t, \sin 2t, t\rangle$ intersect at the origin. Find the angle between $\mathbf{r}_1'(t)$ and $\mathbf{r}_2'(t)$ at $t = 0$. [5]

g. Find the limit, if it exists, or show that the limit does not exist [5]

$$\lim_{(x,y)\to(0,0)} \frac{x^2 + \sin^2 y}{2x^2 + y^2}.$$

**Question 2 (30 points).**

a. Find the sum of the convergent series below by using a well-known function. Identify the function and explain how you obtained the sum, [5]

$$\sum_{n=0}^{\infty} \frac{1}{2^n n!}.$$

b. It can be shown that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

Use this fact to show that [10]

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} = \frac{\pi^2}{8}.$$

c. Find the exact length of the curve [5]

$$x = e^t \cos t, \quad y = e^t \sin t, \quad 0 \leq t \leq \pi.$$

d. Find the gradient of the function $f(x, y, z) = \sqrt{x + yz}$. [5]

e. Use implicit differentiation to find $\partial z/\partial x$ and $\partial z/\partial y$ [5]

$$x^2 - y^2 + z^2 - 2z = 4.$$

**Question 3 (35 points).**

a. Find the area of the region that lies inside the first curve and outside the second curve [10]

$$r = 1 - \sin\theta, \quad r = 1.$$

b. Use Lagrange multipliers to find the maximum and minimum values of $f$ subject to the given constraint [10]

$$f(x, y) = x^2 y, \quad x^2 + y^2 = 1.$$

c. Find the equation of the tangent plane to the given surface at the specified point [5]

$$F(x, y, z) = y^2 + z^2 - x, \quad P(3, 1, -1).$$

d. Let $p(t) = f(g(t), h(t))$, where $f$ is differentiable, and $g(2) = 4, g'(2) = -3, h(2) = 5, h'(2) = 6, f_x(4, 5) = 2, f_y(4, 5) = 8$.
Find $p'(2)$. [10]

**-END OF PAPER-**